

Jyotibroto Roy Laskar

Software Engineer Bengaluru, Karnataka, India +91 9832215243

jjyotibrotoroylaskar@gmail.com github.com/JyotibrotoRoy linkedin.com/in/jyotibroto-roy-laskar

Education

Presidency University, Bengaluru - BCA (Data Science) CGPA: 8.7/ 10 August 2023 to Present

Project Experience

Auradrive - Real time Ride Hailing Backend July 2025 to September 2025
API Documentation
GitHub

- Built a **cloud-native backend** with Spring Boot and MongoDB, securing **multi-role APIs** using Spring Security and JWT.
- Designed a **real-time driver matching system** using Redis Geospatial indices, achieving sub-second response times.
- Implemented **distributed locking** to prevent race conditions during simultaneous booking requests.
- Implemented real-time trip updates and instant OTP delivery using Spring WebSockets, ensuring seamless synchronization between drivers and riders.
- Deployed a **dockerized production system** on Render with **AWS S3** for secure document storage.

Cloud Infrastructure Observability & Performance Dashboard August 2025 to Present
Live Application
GitHub

- Built a **real-time observability platform** tracking p99 latency, error rates, and memory usage across distributed systems.
- Implemented a **real-time data ingestion pipeline** using **WebSockets and Supabase Realtime** to stream **time-series performance metrics**, reducing observability lag to **under 500 milliseconds**.
- Built **interactive dashboards and telemetry visualizations** using **Recharts and D3.js** to analyze correlations between **memory consumption and request latency** during peak traffic.
- Optimized **dashboard performance and query execution** by implementing **database indexing and materialized views**, reducing dashboard load times by **60%**.

Bengaluru Raksha - Public Safety Platform (Personal Project) May 2025 to June 2025
Live Demo
GitHub

- Developed a **public safety backend** using **Express.js and MongoDB**, building **RESTful APIs** to manage SOS alerts, emergency contacts, and sensitive user data with high availability.
- Implemented **JWT-based authentication and authorization** and **GPS based SMS alerts** using the **Twilio API**, enabling **secure, low-latency emergency notifications**.
- Optimized **SOS alert and contact lookup flows** by **indexing frequently queried fields**, reducing emergency request response times under peak usage.

Technologies

Languages: Java, C/C++, JavaScript, Python

Frameworks : Springboot, Spring Security, Node.js, Express.js, React

Cloud/Database/Tech Stack : MongoDB, PostgreSQL, MySQL, Redis, Hibernate, Spring Data JPA, Spring Websockets, REST APIs, JWT, supabase, Twilio, Multer

Tools & Platforms: Docker, Git, Postman, S3, Swagger